# Justifying constructions using congruent triangles

**ANSWER KEY**

## Core Concepts: Justifying constructions using congruent triangles

**1** In $\triangle ABC$ studying justifying constructions using congruent triangles, $m\angle A = 35^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

**a.** $100^\circ$ ✓

**b.** $110^\circ$

**c.** $92^\circ$

**d.** $145^\circ$

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 45^\circ) = 100^\circ$.

**2** A geometric figure scaled by factor $k = 2$ has an original side length of $4$ cm. What is the corresponding image length?

**a.** 6 cm

**b.** 8 cm ✓

**c.** 12 cm

**d.** 16 cm

Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

**3** On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its $y$-coordinate when $x = 1$?

**a.** 5

**b.** 11

**c.** 9 ✓

**d.** 13

Substituting $x = 1$ into $y = 4x + 5$ gives $y = 4(1) + 5 = 9$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

## Procedural Fluency & Algebraic Manipulation

**4** In $\triangle ABC$ studying justifying constructions using congruent triangles, $m\angle A = 35^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

**a.** $93^\circ$ ✓

**b.** $103^\circ$

**c.** $85^\circ$

**d.** $145^\circ$

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 52^\circ) = 93^\circ$.

**5** A geometric figure scaled by factor $k = 3$ has an original side length of $4$ cm. What is the corresponding image length?

**a.** 9 cm

**b.** 12 cm ✓

**c.** 16 cm

**d.** 36 cm

Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

**6** On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its $y$-coordinate when $x = 2$?

**a.** 9

**b.** 15

**c.** 13 ✓

**d.** 17

Substituting $x = 2$ into $y = 4x + 5$ gives $y = 4(2) + 5 = 13$.

---

## Graphical & Geometric Interpretation

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**7** In $\triangle ABC$ studying justifying constructions using congruent triangles, $m\angle A = 35^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

**a.** $86^\circ$ ✓

**b.** $96^\circ$

**c.** $78^\circ$

**d.** $145^\circ$

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 59^\circ) = 86^\circ$.

**8** A geometric figure scaled by factor $k = 4$ has an original side length of $4$ cm. What is the corresponding image length?

**a.** 12 cm

**b.** 16 cm ✓

**c.** 20 cm

**d.** 64 cm

Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

**9** On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its $y$-coordinate when $x = 3$?

**a.** 13

**b.** 19

**c.** 17 ✓

**d.** 21

Substituting $x = 3$ into $y = 4x + 5$ gives $y = 4(3) + 5 = 17$.

---

## Exam-Style Problem Solving & Modeling

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

**10** In $\triangle ABC$ studying justifying constructions using congruent triangles, $m\angle A = 35^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

**a.** $79^\circ$ ✓

**b.** $89^\circ$

**c.** $71^\circ$

**d.** $145^\circ$

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (35^\circ + 66^\circ) = 79^\circ$.

**11** A geometric figure scaled by factor $k = 5$ has an original side length of $4$ cm. What is the corresponding image length?

**a.** 15 cm

**b.** 20 cm ✓

**c.** 24 cm

**d.** 100 cm

Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

**12** On the coordinate plane, a line with slope $m = 4$ passes through $(0, 5)$. What is its $y$-coordinate when $x = 4$?

**a.** 17

**b.** 23

**c.** 21 ✓

**d.** 25

Substituting $x = 4$ into $y = 4x + 5$ gives $y = 4(4) + 5 = 21$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647